

1) Bully algorithm for distributed file system.

Ans) 1) Need of the algorithm for the particular system.

A distributed file system consists of multiple nodes that cooperate to store, manage, & access files across a network.

- Managing file metadata.
- Coordinating read/write access.
- Handling lock management
- Monitoring node failures

Importance

- It ensures continuous availability of distributed file system.
- It enables automatic leader election without centralized control.
- It supports fault detection & recovery.

2) Description

It is a leader election algorithm used in distributed systems where:

- each process has unique numeric identifier
- process with highest ID is chosen as coordinator.
- all process know IDs of other processes.

Working

- When a process detects that the coordinator has failed, it initiates an election.
- Process sends an election message to all process with higher IDs.
- If no response is received within a timeperiod, the process declares itself the coordinator.

- If a higher ID process responds with OK message, then it takes over.
- Eventually, the highest ID process wins & sends a coordinator message to all lower ID processes.
- Then others update their coordinator info accordingly.

3) Real-time application

- Distributed File Systems
- Distributed Databases
- Cluster Management Systems
- Cloud Computing Platforms

4) Requirements

a) System req.

- Each node must have unique identifiers.
- All nodes must know ID of other nodes.
- Failure detection mechanism.

b) Software req.

- Message handling for election, ok, & coordinator messages.
- Timeout management.
- Process state "
- Error handling.

Network req.

- Point-to-point communication b/w nodes.
- Reliable delivery of messages.
- Support concurrent messaging.

Conclusion

The Bully Algorithm plays a crucial role in ensuring reliability, fault tolerance, & coordination in a distributed file system. It helps maintain system consistency & availability. Although it has higher message overhead, its simplicity & fast leader recovery make it effective for many distributed envs.